

1. Overview

Phase 1 — Outcome Distribution: What is the historical distribution of career outcomes for a given class of player (e.g., a 3-star recruit, a Power 4 transfer)?

Phase 2 — Value Correlation: How do those outcome distributions translate into expected value, and how does a proposed compensation package compare against that value?

2. Users and Use Cases

2.1 Primary User

Stanford Football front office staff and coaches evaluating whether to offer a prospective or current player a specific financial package.

2.2 Core Use Cases

- A recruiter is evaluating a 4-star high school QB. They want to see, historically, what percentage of 4-star QBs became starters vs. busts, and how that distribution shifts across Years 1–5.
- The GM is deciding between two transfer portal targets at different price points. They want to compare the expected value and downside risk of each option against the proposed compensation.
- The front office is building a class budget. They want to stress-test the total expected return of a group of signings under best-case, median, and worst-case scenarios.

3. Requirements

3.1 Phase 1: Outcome Distribution

The goal of Phase 1 is to answer: “For a player with profile X, what does the historical distribution of career outcomes look like?”

3.1.1 Outcome Categories

Attempting to group all players into the following categories:

Outcome	Definition
Bust	Minimal or no meaningful on-field contribution during career at the program.

Depth / Rotation	Contributes as a backup or rotational player; not a consistent starter.
Starter	Earns and holds a starting role for at least one full season.
Impact Player	High-level starter: All-Conference caliber, NFL draft pick, or equivalent on-field impact.

3.1.2 Player Input Profile

The system must accept the following inputs to define a player's profile for lookup or projection:

Input Field	Type	Notes
Recruiting Star Rating	Integer	247Sports composite
Position Group	Categorical	QB, RB, WR, TE, OL, DL, LB, DB, Special Teams.
Player Source	Categorical	High school recruit vs. transfer portal entry.
Composite Rating	Float	247Sports composite numeric rating for finer-grained cohort matching.
Year in Program	Integer	Which year of eligibility the projection covers.

3.1.3 Output: Probability Distribution

For a given player profile, the system must return a probability distribution across the four outcome categories for each year of eligibility (Years 1–5).

Example output for a 3-star WR recruit:

Outcome	Year 1	Year 2	Year 3	Year 4	Year 5
Bust	50%	42%	35%	32%	32%
Depth	35%	33%	30%	28%	25%
Starter	12%	18%	25%	28%	28%
Impact	3%	7%	10%	12%	15%

Probabilities for each year must sum to 100%. The system should display these as both a table and a stacked bar or area chart.

3.1.4 Data Requirements

Phase 1 uses publicly available data.

Data Source	Key Fields	Priority
247Sports Composite Rankings	Star rating, numeric composite, position, recruit class year	P0
Public Roster / Participation Data	Player name, school, years on roster, starts, depth chart position	P0
Transfer Portal Rankings	Star rating at transfer, origin school, destination, position	P1
Draft Outcomes	Round drafted, undrafted, position	P1
PFF Grades	Overall grade, position grade, snap counts	P1

The dataset should cover at minimum the last 10 recruiting classes (2015–2024) across Power 4 / Power 5 conferences to build meaningful cohorts. Data should be scoped to the ACC or Power 4 context wherever possible.

3.1.5 Modeling Approach

Flexibility in modeling approach. Some ideas include:

- Historical cohort analysis: group players by star rating, position, and source, then compute empirical outcome distributions.
- Bayesian modeling: use priors from historical cohorts and update with any player-specific information.
- Logistic regression or multinomial classification: predict outcome category probabilities from input features.

For the PoC, a cohort-based approach is likely sufficient.

3.1.6 Phase 1 Deliverables

- Data pipeline (scripts or notebooks) that ingests, cleans, and stores historical player data.
- A model that accepts a player profile and returns a year-by-year outcome probability distribution.
- (Nice to Have) An app that lets a user select a player profile and view the resulting distribution table and chart.
- Documentation: data sources, assumptions, known limitations, and instructions to reproduce.

3.2 Phase 2: Value Correlation and Investment Evaluation

Phase 2 layers financial evaluation on top of the outcome distributions from Phase 1. The goal is to answer: “Given this player’s projected outcome distribution, is offering them \$X a good investment?”

3.2.1 Value Mapping

We will need an external source to provide a value mapping table that assigns a dollar value to each outcome category, by position. The system must ingest this mapping and apply it to the Phase 1 probability distributions.

Sample mapping (to be overridden by future actuals):

Outcome	Default Value
Bust	\$0
Depth / Rotation	\$100,000
Starter	\$400,000
Impact Player	\$1,200,000

The mapping should be configurable by position group (e.g., an Impact QB may be valued differently than an Impact OL). The system should store this as an editable config file or database table.

3.2.2 Investment Evaluation Metrics

Given a player’s outcome distribution (from Phase 1), the value mapping, and a user-entered proposed compensation amount, the system must compute:

Metric	Calculation	Purpose
Expected Value (EV)	$\sum (P(\text{outcome}) \times \text{Value}(\text{outcome}))$ across all years	What is this player likely worth?
Net Value	EV – Total Compensation	Are we paying more or less than expected value?
ROI	EV / Total Compensation	Efficiency of the investment.
Downside Probability	P(Bust) across career	How likely is total loss?

Upside Probability	P(Impact) across career	How likely is a home run?
Outcome Variance	Var(Value) from distribution	How spread out are the possible outcomes?

3.2.3 Monte Carlo Simulation

The system should support Monte Carlo simulation to generate a distribution of possible total career value outcomes for a given player. The simulation should:

- Run N iterations, sampling from the year-by-year outcome probabilities.
- For each iteration, sample one outcome per year, look up the value, and sum across the career.
- Output a histogram of total career value outcomes.
- Display summary statistics: median, mean, 10th percentile (downside), 90th percentile (upside), and the proposed compensation as a reference line.

3.2.4 Comparison Mode

The UI should support comparing two players side-by-side. A recruiter should be able to input Player A (profile + comp) and Player B (profile + comp) and see their EV, Net Value, ROI, and risk metrics in a comparison table and overlaid distribution charts.

3.2.5 Phase 2 Deliverables

- A value mapping module that accepts a player value table and integrates it with Phase 1 output.
- An evaluation model that computes EV, Net Value, ROI, and risk metrics for a given player + compensation input.
- A Monte Carlo simulation module that outputs a distribution of career value outcomes.
- Extended UI with: compensation input, metric dashboard, simulation histogram, and player comparison view.
- Updated documentation covering valuation methodology and simulation assumptions.

4. Success Criteria

4.1 Phase 1

- A user can select a player profile (star rating, position, source)

- The app displays a year-by-year probability distribution table and stacked bar chart.
- Distributions are derived from real historical data (not hardcoded).
- The dataset covers at least 5 recruiting classes and 3+ position groups.
- Code is modular, documented, and reproducible from a clean clone.

4.2 Phase 2

- A user can enter a compensation amount and see EV, Net Value, ROI, and risk metrics.
- Monte Carlo simulation runs and displays a histogram with summary stats.
- Two players can be compared side-by-side.
- Value mappings are configurable without code changes.

5. Suggested Timeline

Milestone	Target	Description
Phase 1: Data Pipeline	Weeks 1–2	Scrape / ingest recruiting + roster data. Clean and store.
Phase 1: Model v1	Weeks 3–4	Build cohort-based outcome distributions. Validate against known outcomes.
Phase 1: App MVP	Week 5	Wire up UI to model. Ship Phase 1 for review.
Phase 2: Value + Eval Engine	Weeks 6–7	Integrate value mapping. Compute EV, ROI, risk metrics.
Phase 2: Simulation	Week 8	Monte Carlo module + histogram visualization.
Phase 2: Comparison + Polish	Weeks 9–10	Player comparison mode. Final QA and documentation.

6. Risks and Assumptions

Item	Type	Mitigation
Small cohort sizes for niche profiles (e.g., 5-star TEs)	Risk	Fall back to broader cohorts (e.g., all 5-stars, or all TEs) and note reduced confidence.
Public data may lack reliable “starter” classification	Risk	Define starter heuristics clearly (e.g., ≥ 10 starts in a season). Document the definition.

247Sports data may require scraping	Risk	Use respectful scraping with caching. Explore APIs or pre-compiled datasets first.
Value mappings are placeholders	Assumption	Front office will provide real values. System must make these easily swappable.

8. Out of Scope (for PoC)

- Portfolio-level roster optimization (evaluating an entire class together).
- Real-time data feeds or automated updates.
- Authentication, multi-user access, or deployment to production infrastructure.
- Modeling transfer likelihood, injury risk, or attrition.
- Integration with internal recruiting database.